

Q3

Given: $m = 20.2\text{u}$

$$= 20.2 \times 1.66 \times 10^{-27} \text{ kg} = 3.35 \times 10^{-26} \text{ kg};$$

$M_s = 12.0 \text{ kg}; V = 8.0 \times 10^{-3} \text{ m}^3;$

$n = 4.50 \text{ mol}; T = 300 \text{ K};$

Specific heat capacity of steel

$$c_s = 448 \text{ J kg}^{-1} \text{ K}^{-1}$$

Molar heat capacity of neon

$$= C_n = 12.5 \text{ J mol}^{-1} \text{ K}^{-1}$$

Specific heat capacity of water

$$c_w = 4186 \text{ J kg}^{-1} \text{ K}^{-1}$$

Specific latent heat of fusion of water

$$l_f = 3.33 \times 10^5 \text{ J kg}^{-1}$$

Density of water $\rho_w = 1.00 \times 10^3 \text{ kg m}^{-3}$

Density of ice $\rho_i = 917 \text{ kg m}^{-3}$

3(a)

Using the Ideal Gas Equation,

$$p = \frac{nRT}{V}$$
$$p = \frac{(4.50)(8.31)(300)}{8.0 \times 10^{-3}}$$
$$\mathbf{p = 1.40 \times 10^6 \text{ Pa}}$$

3(b)

Using the Pressure Law for constant mass of gas and volume, we have

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
$$p_2 = \left(\frac{1.40 \times 10^6}{300} \right) 273$$
$$\mathbf{p_2 = 1.27 \times 10^6 \text{ Pa}}$$

3(c)

Loss of energy of neon gas + loss of energy by steel canister = gain in energy by ice in melting

$$nC_n(300 - 273) + M_sc_s(300 - 273) = M_ili_f$$
$$M_i = \frac{(4.50)(12.5)(27) + (12.0)(448)(27)}{3.33 \times 10^5}$$
$$\mathbf{= 0.440 \text{ kg}}$$

4(a)(i)

Diffusion of liquids is faster than diffusion of solids and slower than gases.